

50.007 Machine Learning, Spring 2025
Homework 5 (Practice)

Sample Solutions

In this homework, we would like to look at some simple MDP problem.

Consider the Markov decision process (MDP) associated with a simplified version of the robot that we showed during class that plays the guessing game with us. It has 4 states $\{Uncertain, Certain, Lose, Win\}$, with Win being the final state. The following table summarizes the actions that the robot can take at each state, and the transition probabilities from one state to another after taking a certain action. For example, when the robot is at the $Certain$ state (it is certain about the answer), there are two possible actions to take: ask (A) and guess (G). If it takes the action G, then there is a probability 0.7 that the robot will arrive at the Win state (and wins the game).

s	a	s'	$T(s, a, s')$
$Uncertain$	A	$Uncertain$	0.9
$Uncertain$	A	$Certain$	0.1
$Uncertain$	G	$Lose$	0.9
$Uncertain$	G	Win	0.1
$Certain$	A	$Certain$	1.0
$Certain$	G	$Lose$	0.3
$Certain$	G	Win	0.7
$Lose$	A	$Uncertain$	0.8
$Lose$	A	$Certain$	0.2
$Lose$	G	$Lose$	0.8
$Lose$	G	Win	0.2
Win	A	Win	1.0
Win	G	Win	1.0

The reward function $R(s, a, s') = R(s')$ is defined as:

s'	$Uncertain$	$Certain$	$Lose$	Win
$R(s')$	0.0	1.0	-2.0	2.0

The discount factor is $\gamma = 0.5$.

1. Suppose we initialize $Q_0^*(s, a) = 0$ for all $s \in S$ and $a \in \{A, G\}$. Evaluate the Q-values $Q_1^*(s, a)$ after exactly one iteration of the Q-Value Iteration Algorithm (assume we perform synchronized updates. In other words, we always use Q_0 values when we calculate Q_1 values). Write your answers in the table below.

	$s = Uncertain$	$s = Certain$	$s = Lose$	$s = Win$
A	0.1	1.0	0.2	2.0
G	-1.6	0.8	-1.2	2.0

2. What is the policy that we would derive from $Q_1^*(s, a)$? Answer by filling in the action that should be taken at each state in the table below (in case of draw, the action G is preferred).

	$s = \text{Uncertain}$	$s = \text{Certain}$	$s = \text{Lose}$	$s = \text{Win}$
	A	A	A	G

3. What are the values $V_1^*(s)$ corresponding to $Q_1^*(s, a)$?

	$s = \text{Uncertain}$	$s = \text{Certain}$	$s = \text{Lose}$	$s = \text{Win}$
	0.1	1.0	0.2	2.0

4. Evaluate the Q-values $Q_2^*(s, a)$ after exactly two iterations of the Q-Value Iteration Algorithm. Write your answers in the table below.

	$s = \text{Uncertain}$	$s = \text{Certain}$	$s = \text{Lose}$	$s = \text{Win}$
A	0.195	1.5	0.34	3.0
G	-1.41	1.53	-0.92	3.0

5. What is the policy that we would derive from $Q_2^*(s, a)$? Answer by filling in the action that should be taken at each state in the table below (in case of draw, the action G is preferred).

	$s = \text{Uncertain}$	$s = \text{Certain}$	$s = \text{Lose}$	$s = \text{Win}$
	A	G	A	G

6. What are the values $V_2^*(s)$ corresponding to $Q_2^*(s, a)$?

	$s = \text{Uncertain}$	$s = \text{Certain}$	$s = \text{Lose}$	$s = \text{Win}$
	0.195	1.53	0.34	3.0